

1. Scope

1.1 Purpose

This document defines verifiable requirements for a CubeSat class Attitude Determination and Control System simulation and the verification methods used to demonstrate control compliance.

1.2 System Under Test

The system under test is a rigid body spacecraft attitude simulation with:

- Attitude state as scalar first quaternion and body rates
- Reaction wheels and thruster based external torque actuation for wheel desaturation
- Gyroscope and star tracker sensors with sampling and noise models
- Estimator using an error state EKF attitude filter
- Controller for detumble, attitude tracking, and desaturation management
- Environmental disturbance torque models implemented as direct applied torque inputs

1.3 Definitions

- Attitude error: principal rotation angle between commanded and estimated attitudes
 - Settling time: time to enter and remain within an error bound
 - RMS error: root mean square attitude error over a specified interval
 - NIS: normalized innovation squared for star tracker update residuals
 - Desaturation mode: control mode that uses external torque to reduce reaction wheel momentum while maintaining bounded attitude error
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2. Assumptions and Simulation Conditions

2.1 Reference Conditions

Unless otherwise specified, verification runs shall use:

- Base step T_s base 0.01 s
- Gyro sample T_s gyro 0.01 s
- Star tracker sample T_s star 0.2 s
- Wheel torque limit per axis $\tau_{\text{wheel max}}$ 0.01 N m
- Wheel max speed $\Omega_{\text{wheel max}}$ 6000 rpm equivalent
- Disturbance torque baseline magnitude $1e-5$ N m per axis

- Thruster torque limit per axis $\tau_{thr\ max}$ as configured in the simulation configuration

2.2 Coordinate and Sign Conventions

- Quaternion is scalar first and unit norm
- Body rates are expressed in the body frame
- Commanded body torque is in the body frame
- Thruster generated torque is applied in the body frame

2.3 Disturbance Modeling Scope

- The simulation shall not include gravity gradient, aerodynamic torque, solar radiation pressure torque, or magnetic disturbance torque as separate computed environment models
 - Environmental effects shall be represented by configurable direct disturbance torque inputs, including constant and time varying components
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3. Requirements

Verification methods are coded as:

- A Analysis
- T Test
- I Inspection
- D Demonstration

3.1 Safe Mode and Detumble

ADCS-1: Detumble Rate Reduction

- Statement: The system shall reduce the body rate magnitude from an initial condition of up to 10 deg per second to less than or equal to 0.5 deg per second within 120 s with reaction wheels enabled and zero commanded attitude
- Verification: T
- Verification evidence: Time history of body rate magnitude, settling time measurement

ADCS-2: Detumble Stability Under Saturation

- Statement: The system shall remain stable during detumble when wheel torque commands saturate at $\tau_{wheel\ max}$, without divergence in attitude state or estimator covariance
- Verification: T
- Verification evidence: Bounded quaternion norm, bounded covariance diagonals, no NaN or Inf events

ADCS-3: Wheel Speed Safety During Detumble

- Statement: The system shall maintain wheel speeds within plus or minus Omega wheel max during detumble for the test case in ADCS-1
 - Verification: T
 - Verification evidence: Wheel speed time histories, max absolute wheel speed
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3.2 Attitude Control and Tracking

ADCS-4: Coarse Pointing Acquisition

- Statement: The system shall achieve an attitude error less than or equal to 0.5 deg within 10 s after detumble completion for a fixed commanded attitude
- Verification: T
- Verification evidence: Attitude error time history, acquisition time measurement

ADCS-5: Fine Pointing Hold Baseline

- Statement: The system shall maintain RMS attitude error less than or equal to 0.05 deg over a continuous 60 s hold interval under baseline disturbance torques
- Verification: T
- Verification evidence: RMS attitude error computed over the hold interval

ADCS-6: Slew and Settle Performance

- Statement: The system shall execute a 90 deg step command about each principal axis and settle to within 0.5 deg within 10 s and to within 0.05 deg within 30 s for each axis test
- Verification: T
- Verification evidence: Attitude error time histories per axis, settling time calculations

ADCS-7: Overshoot Control

- Statement: The system shall demonstrate less than or equal to 10 percent overshoot in attitude error for the maneuver in ADCS-6
 - Verification: T
 - Verification evidence: Peak attitude error relative to final bound
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3.3 Disturbance Modeling and Rejection

ADCS-8: Constant Disturbance Rejection

- Statement: The system shall maintain RMS attitude error less than or equal to 0.05 deg for a 60 s hold interval with a constant disturbance torque of magnitude 1×10^{-5} N m applied on each axis
- Verification: T
- Verification evidence: RMS attitude error during hold, disturbance torque configuration log

ADCS-9: Time Varying Disturbance Rejection

- Statement: The system shall remain stable and maintain bounded wheel speeds with a sinusoidal disturbance torque of amplitude 1×10^{-5} N m and frequency 0.05 Hz applied on each axis during a 90 deg slew and hold sequence
- Verification: T
- Verification evidence: Attitude and wheel speed time histories, stability confirmation

ADCS-10: Disturbance Torque Configurability

- Statement: The simulation shall provide a configurable disturbance torque subsystem that supports at minimum constant and sinusoidal torque components with user set amplitude, frequency, and per axis values
- Verification: I
- Verification evidence: Model inspection, parameter list, subsystem interface check

ADCS-11: Disturbance Torque Enablement

- Statement: The disturbance torque subsystem shall be enableable and disableable via configuration flags and shall log its enabled state per run
- Verification: T
- Verification evidence: Configuration log, on versus off comparison run showing nonzero versus zero applied disturbance torque

Note: Requirements ADCS-10 through ADCS-13 from the previous document are removed because gravity gradient, SRP, aerodynamic, and magnetic disturbance components will not be computed as separate models in this simulation scope.

3.4 Momentum Management and Desaturation

ADCS-12: Wheel Momentum Buildup Monitoring

- Statement: The system shall compute and log wheel angular momentum and wheel speed margins continuously during all simulation modes
- Verification: I

- Verification evidence: Logged signals list, unit consistency check

ADCS-13: Thruster Based Desaturation Capability

- Statement: The system shall include a thruster torque subsystem capable of reducing wheel speeds to within 20 percent of zero speed within 300 s during a desaturation mode while maintaining attitude error less than or equal to 2 deg
- Verification: T
- Verification evidence: Wheel speed decay plots, attitude error plots during desaturation

ADCS-14: Thruster Torque Command Saturation

- Statement: The thruster torque subsystem shall include torque command saturation consistent with configurable per axis maximum torque limits
- Verification: I
- Verification evidence: Model inspection, saturation parameter, command clamping check

ADCS-15: Desaturation Mode Threshold Logic

- Statement: The system shall enter desaturation mode when any reaction wheel speed magnitude exceeds 80 percent of Omega wheel max and shall exit desaturation mode only after all wheel speed magnitudes fall below 50 percent of Omega wheel max
- Verification: T
- Verification evidence: Mode flag log, wheel speed time histories, threshold crossing times

ADCS-16: Desaturation Mode Stability

- Statement: During desaturation mode, the system shall maintain bounded body rates and shall not violate quaternion unit norm constraints
- Verification: T
- Verification evidence: Body rate time histories, quaternion norm time history, max deviation

ADCS-17: Thruster Pulse Modulation Behavior

- Statement: If thruster commands are modeled as on off actuation, the simulation shall enforce a configurable minimum pulse on time and shall log commanded pulse events
- Verification: I and T
- Verification evidence: Model inspection, pulse event log, sample time comparison showing pulse enforcement

3.5 Estimation Performance

ADCS-18: Attitude Estimate Boundedness

- Statement: The attitude estimator shall maintain unit norm quaternion output within a tolerance of $1e-6$ for all simulation time
- Verification: T
- Verification evidence: Quaternion norm time history, max deviation

ADCS-19: Star Tracker Update Handling

- Statement: The estimator shall accept star tracker updates at T_s star and ignore updates when star valid is false
- Verification: T
- Verification evidence: Star valid pulse log, innovation updates only on valid pulses

ADCS-20: Star Tracker Dropout Robustness

- Statement: The system shall remain stable with star tracker invalid for 10 s and shall recover to an attitude error less than or equal to 0.5 deg within 5 star tracker updates after validity returns
- Verification: T
- Verification evidence: Attitude error plot across dropout window, recovery time

ADCS-21: Innovation Consistency Monitoring

- Statement: The estimator shall compute and log NIS for star tracker updates and shall flag NIS exceeding a configurable chi square threshold for 3 degrees of freedom
- Verification: T
- Verification evidence: NIS time history, threshold crossing flags

3.6 Control System Robustness

ADCS-22: Closed Loop Stability Margins

- Statement: The control design shall demonstrate gain margin greater than or equal to 6 dB and phase margin greater than or equal to 45 deg for the linearized small angle closed loop model about a nominal attitude hold condition
- Verification: A
- Verification evidence: Linearization operating point definition, Bode plot margins, saved margin report

ADCS-23: Reaction Wheel Friction Modeling

- Statement: The reaction wheel subsystem shall include configurable viscous and Coulomb friction terms affecting wheel acceleration and back torque
- Verification: I
- Verification evidence: Model inspection, parameter list, friction torque equation in documentation

ADCS-24: Actuator Rate Limiting

- Statement: The reaction wheel subsystem shall include configurable torque slew rate limiting and shall log slew limited events
- Verification: I
- Verification evidence: Model inspection, parameter list, event log

3.7 Software Quality and Reproducibility

ADCS-25: Deterministic Monte Carlo Reproducibility

- Statement: The simulation shall produce identical results for a fixed random seed and configuration across repeated runs on the same machine
- Verification: T
- Verification evidence: Hash or checksum of key logged signals, repeated run comparison

ADCS-26: Unit Test Harness Coverage

- Statement: The project shall include unit test harness models for the plant, actuator, sensors, and estimator subsystems with automated pass fail assertions
- Verification: I and T
- Verification evidence: Test list, harness models, test results report

ADCS-27: Configuration Logging

- Statement: Each simulation run shall log a configuration record including parameter values, random seed, version identifier, and enabled subsystem flags
- Verification: T
- Verification evidence: Logged config struct or file output per run

4. Verification Artifacts

For each requirement verified by test, the verification package shall include:

- Plot of relevant time series

- Scalar summary metrics
 - Pass fail statement with requirement ID
 - Stored run configuration and seed
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5. Traceability Matrix

A traceability matrix shall map each requirement ID to:

- Simulink subsystem
 - Test harness name
 - Script or test file name
 - Logged signals used for verification
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6.0 Test List

T-001 Safe Mode Detumble Baseline

Verifies: ADCS-1, ADCS-2, ADCS-3

Run the sim with reaction wheels enabled, zero attitude command, and an initial rate up to the requirement limit. Log body rate magnitude, wheel torque commands, wheel speeds, quaternion norm, and estimator covariance diagonals. Pass if body rates fall below the threshold within the time limit, wheel speeds stay within limits, and no states or covariances diverge.

T-002 Detumble With Torque Saturation Stress

Verifies: ADCS-2, ADCS-3

Force saturation by either increasing initial rates or lowering tau wheel max while keeping the same controller. Verify stability under saturation: quaternion remains normalized, rates remain bounded, covariance stays finite, and the simulation completes without NaN or Inf.

T-003 Coarse Acquisition After Detumble

Verifies: ADCS-4

Run detumble to completion, then apply a fixed commanded attitude. Measure the time from detumble completion to entering and staying within the coarse pointing bound. Log attitude error angle, rates, and actuator usage.

T-004 Fine Pointing Hold With Constant Disturbance

Verifies: ADCS-5, ADCS-8, ADCS-12

Enable constant disturbance torque on all axes at the baseline magnitude and command a fixed attitude hold. Compute RMS attitude error over a continuous 60 s window after transients. Log wheel speeds and wheel angular momentum to show bounded momentum and margin.

T-005 Slew 90 Deg Step About X

Verifies: ADCS-6, ADCS-7

Command a 90 deg step about the X axis starting from a stable hold. Measure settle times to the two error bounds and compute overshoot. Log wheel torques and speeds to confirm the slew does not violate actuator constraints.

T-006 Slew 90 Deg Step About Y

Verifies: ADCS-6, ADCS-7

Same as T-005 but with a 90 deg step about Y.

T-007 Slew 90 Deg Step About Z

Verifies: ADCS-6, ADCS-7

Same as T-005 but with a 90 deg step about Z.

T-008 Time Varying Disturbance During Slew And Hold

Verifies: ADCS-9, ADCS-3

Apply a sinusoidal disturbance torque at the defined amplitude and frequency while running a slew and subsequent hold. Confirm stability, bounded wheel speeds, and no runaway attitude error.

T-009 Disturbance Torque Subsystem Enablement And Config Test

Verifies: ADCS-10, ADCS-11

Inspect the disturbance subsystem to confirm constant and sinusoidal modes exist and are parameterized. Run an on versus off test and confirm the applied disturbance torque is nonzero when enabled and zero when disabled. Log enabled flags and parameters.

T-010 Wheel Momentum Monitoring And Logging Audit

Verifies: ADCS-12

Audit logged signals and confirm wheel speeds, wheel angular momentum, and margin to Omega wheel max are recorded for every run. Include one run to prove signals are populated and nontrivial.

T-011 Thruster Desaturation Mode Functional Test

Verifies: ADCS-13, ADCS-14, ADCS-15, ADCS-16

Start with wheels spun up near a high fraction of Omega wheel max and enter desaturation mode. Command thruster torque to dump momentum and show wheel speeds reduce to within the specified band within the time limit while attitude error stays within the allowed bound. Verify thruster torque commands saturate at the configured maximum and that saturation events are logged.

T-012 Estimator Quaternion Norm Integrity

Verifies: ADCS-18

Run a representative scenario including slews and holds and log the estimator quaternion norm. Pass if norm error stays within tolerance for the full run.

T-013 Star Tracker Valid Pulse Handling

Verifies: ADCS-19

Create a test where star valid toggles false for controlled intervals while the star tracker measurement continues to update. Verify the estimator performs measurement updates only when star valid is true.

T-014 Star Tracker Dropout Robustness

Verifies: ADCS-20

Disable star valid for a full 10 s and then re enable it. Verify stability and recovery to the error bound within the required number of star tracker updates.

T-015 Innovation And NIS Monitoring With Gating

Verifies: ADCS-21

Inject a small number of corrupted star tracker measurements or noise bursts. Verify NIS is computed and that threshold exceedances are flagged and optionally rejected.

T-016 Linearized Stability Margins Report

Verifies: ADCS-22

Linearize the closed loop system about a nominal attitude hold operating point and compute gain and phase margins. Save the margin plot outputs and the operating point definition used to generate them.

T-017 Reaction Wheel Friction Model Unit Test

Verifies: ADCS-23

With a fixed commanded torque and no external spacecraft motion, validate wheel acceleration and steady state behavior under viscous and Coulomb friction settings. Include a comparison plot with friction off and on.

T-018 Actuator Torque Slew Rate Limiting Test

Verifies: ADCS-24

Command a step in desired torque and verify the actuator output ramps at the configured slew rate rather than stepping instantly. Confirm slew limiting events are logged.

T-019 Deterministic Reproducibility Test

Verifies: ADCS-25

Run the same simulation twice with the same random seed and configuration and confirm key logged signals match exactly.

T-020 Unit Test Harness Coverage Audit

Verifies: ADCS-26

Confirm harnesses exist for plant, actuator, sensors, and estimator with pass fail assertions and an automated script to run them.

T-021 Configuration Logging Validation

Verifies: ADCS-27

Validate that each run logs a configuration record including parameters, seed, version identifier, and enabled subsystem flags, and that rerunning from the config reproduces results.